The New Classical Approach

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1 Introduction

The New Classical approach extends monetarist skepticism about policy fine-tuning by sharpening the mechanism through which policy affects real outcomes. The key shift is from short-run money illusion under adaptive expectations (Friedman's framework) to model-consistent forecasting under rational expectations, which removes systematic policy surprises and eliminates systematic real effects of monetary policy.

1.1 Expectations and the rational expectations revolution

1.1.1 Adaptive expectations

Adaptive expectations operate as a trial-and-error forecasting rule where agents update forecasts using past observed outcomes:

Price expectations: $E_{t-1}P_t = f(P_{t-1}, P_{t-2}, ...)$

A common error-correction form is:

$$\pi_t^e = \pi_{t-1}^e + \lambda(\pi_{t-1} - \pi_{t-1}^e), \quad 0 < \lambda \leq 1.$$

Adaptive expectations permit systematic forecast errors during regime changes, creating temporary money illusion and thus temporary real effects.

1.1.2 Rational expectations

Rational expectations are defined by three criteria: (i) forecasts use the true economic model (agents know the economy's structure), (ii) forecasts condition on all available information, and (iii) forecast errors are purely random—systematic errors are eliminated.

In classical formulation: expectations are "the same as the predictions of the relevant economic theory," and the economy does not waste information.

Model-consistent forecasting:

$$E_{t-1}P_t = E(P_t \mid I_{t-1}),$$

where I_{t-1} is the information set at time $t-1$.

If workers have access to unbiased forecasts, predictable policy loses its ability to manipulate perceived real wages and thus loses its ability to shift real output, even in the short run.

2 The New Classical model of output and prices

2.1 Lucas aggregate supply and aggregate demand

The New Classical model consists of two equations: Lucas supply (LAS) and aggregate demand (AD).

Lucas aggregate supply (LAS): Output deviates from potential only when the price level differs from expectations:

$$Y_t^s - Y^* = \kappa(P_t - E_{t-1}P_t).$$

Lucas (1972) derives this in a market-clearing framework with optimizing agents.

Interpretation: - If $P_t > E_{t-1}P_t$: firms perceive relative-price gains, supply rises ($Y_t^s > Y^*$) - If $P_t = E_{t-1}P_t$: output at potential ($Y_t^s = Y^*$) - If $P_t < E_{t-1}P_t$: output below potential ($Y_t^s < Y^*$)

The parameter κ measures supply responsiveness: $\partial Y_t/\partial(P_t - E_{t-1}P_t) = \kappa$.

Aggregate demand (AD): In log form:

$$Y_t^d = \gamma + M_t - P_t,$$

where M_t is nominal money and γ captures exogenous demand shifts.

Agents know structural parameters γ, κ, and natural output Y^*.

2.2 Solving the model: the role of κ

Equilibrium: $Y_t^s = Y_t^d = Y_t$. Combining LAS and AD:

1. Under rational expectations:
$$E_{t-1}Y_t = Y^*.$$

2. Output deviations depend only on monetary surprises:
$$Y_t - Y^* = \frac{\kappa}{1+\kappa}(M_t - E_{t-1}M_t).$$

3. Price surprises scale with monetary surprises:
$$P_t - E_{t-1}P_t = \frac{1}{1+\kappa}(M_t - E_{t-1}M_t).$$

Interpretation: larger κ means output responds more, prices respond less; smaller κ reverses this. The Lucas supply slope (P on vertical axis, Y horizontal) is proportional to $1/\kappa$.

When policy is volatile, agents discount price movements as noise ("boy who cried wolf"), lowering effective κ and steepening LAS.

3 Monetary policy implications

3.1 The monetary policy ineffectiveness proposition

With rational expectations and market clearing, *systematic and predictable* monetary policy has **no real effect**, even short-run. Predictable policy is incorporated into expectations, shifting prices, not real outcomes.

Formally, if $M_t = E_{t-1}M_t$ (policy is forecastable):
$$Y_t - Y^* = 0.$$

3.2 Systematic rules versus random disturbances

Sharp distinction: (i) predictable rule-like policy vs. (ii) random disturbances.

Money supply process:

$$M_t = (1 + g_M)M_{t-1} + u_t,$$

where u_t is white noise ($E_{t-1}u_t = 0$).

Then: $E_{t-1}M_t = (1 + g_M)M_{t-1}$, so $M_t - E_{t-1}M_t = u_t$.

Output deviations:

$$Y_t - Y^* = \frac{\kappa}{1+\kappa}u_t.$$

Only randomness affects output. Predictable money growth affects the price level, not real activity.

3.3 Why deliberate unpredictability is not recommended

The model should not justify randomness. Two reasons:

1. Random policies produce recessions and booms equally (negative u_t as likely as positive).
2. High volatility changes behaviour: agents discount price signals, steepening LAS (lower effective κ, higher price response).

This logic explains modern central-bank transparency: policy is more effective when expectations are anchored and communication reduces unnecessary uncertainty.

4 Exceptions and policy extensions

Two important exceptions: monetary policy can stabilise real activity under rational expectations if key assumptions are relaxed.

4.1 Asymmetric information and real shocks

If productivity shocks drive cycles and the central bank forecasts a negative shock before the private sector, it can expand policy in advance to stabilise output (at higher prices).

This fits the New Classical framework because the CB has an informational advantage, creating a price surprise from the public's perspective.

Compatible with real-business-cycle reasoning: technology/productivity shocks drive cycles.

4.2 Long-term nominal wage contracts and stabilisation

In many economies, nominal wages are fixed by multi-year contracts. Even if demand shocks are anticipated, monetary policy can offset them because some nominal wages cannot adjust immediately—rigidity creates stabilisation scope.

This aligns with Fischer (1977): under wage rigidity, monetary rules interact with shock propagation even under rational expectations.

Under symmetric information (no CB informational advantage), stabilisation works best for demand shocks; supply-side stabilisation requires informational asymmetry.

4.3 Lucas critique and credibility

The **Lucas critique**: policy-effect estimates from historical relationships are unreliable if decision rules shift with the regime. This motivates structural, microfounded models.

If agents form expectations rationally, **commitment and credibility** shape outcomes through expectations. Rules, strategy statements, and transparent communication are policy tools.

Time-inconsistency literature: policy is strategic interaction with rational agents. Discretionary optimisation can backfire because promises today are not optimal tomorrow (after expectations set). Central to rules-versus-discretion analysis.

5 Modern evidence and applications

5.1 Measuring policy surprises

New Classical model disturbance: $M_t - E_{t-1}M_t$. Modern empirical work isolates surprise components using high-frequency financial data.

Recent IMF working paper: dataset of monetary policy shocks for 21 advanced and 8 emerging economies (2000–2022), using daily changes in interest rate swap rates around central bank announcements to identify unexpected policy shocks to the rate path. Maps directly to model: unexpected shocks are the empirical analogue of u_t.

5.2 Credibility and anchored expectations in practice

New Classical analysis makes credibility central. Modern CBs frame policy as expectations management.

Federal Reserve: publishing its "Statement on Longer-Run Goals and Monetary Policy Strategy" helps the public understand the framework—matters for transparency, accountability, and effectiveness.

ECB (2021 strategy): symmetric 2% medium-term inflation aim; notes that effective lower bound can require "especially forceful or persistent" measures, including transitory inflation above target.

IMF research: better-anchored expectations → less persistent inflation responses to shocks, reinforcing New Classical emphasis on credibility and expectations formation.

5.3 Policy regime episodes

Bank of Japan: reference timeline documents unconventional regimes from late 1990s onward (ZIRP 1999, QE from 2001). Explicitly lists "policy duration effect (forward guidance)" as part of ZIRP—an expectations-management concept compatible with New Classical theory.

Japan's 2024 framework change: QQE with yield curve control and negative rates judged to have "fulfilled their roles," shifting back to short-rate guidance—regime transition where expectations and credibility matter.

ECB (June 2014): introducing negative deposit facility rate (–0.10% effective June 11, 2014) illustrates how rules change when constraints bind and why expectation formation is central to transmission.

COVID-19 emergency actions: central banks reacted with large-scale measures for market functioning and macro stabilisation, showing regime and expectations logic when short rates are constrained.

5.4 Comparative framework: mechanisms and applications

Mechanism	Prediction	Modern illustration
Systematic, predictable policy	If anticipated, shifts $E_{t-1}P_t$ one-for-one; $Y_t \approx Y^*$	Fed and ECB strategy statements anchor expectations through transparency.

Mechanism	Prediction	Modern illustration
Unanticipated shocks	Only surprise component drives $P_t - E_{t-1}P_t$, hence $Y_t - Y^*$	IMF high-frequency dataset (2000–2022) identifies unexpected policy shocks at announcements.
Policy volatility	Agents discount price signals; LAS steepens (lower effective κ)	CBs prefer predictable reaction functions over crisis-era surprise moves.
Wage contracts	Nominal rigidity reopens stabilisation, even under rational expectations	BoE COVID interventions supported market functioning and prevented tightening.
Informational advantage	CB with better real-shock information can offset systematically	BoJ forward guidance and regime design link to RBC-style productivity shocks.

6 Figures

Figure 1. Lucas aggregate supply: output deviates from potential only when P differs from expected price. If expectations are correct, output remains at potential.

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Figure 2. Predictable versus unpredictable monetary policy. A predictable rise in money shifts AD and LAS upward (via expectations); equilibrium stays at Y^*. An unpredictable rise shifts AD without shifting LAS initially; equilibrium moves along LAS with Y above Y^*.

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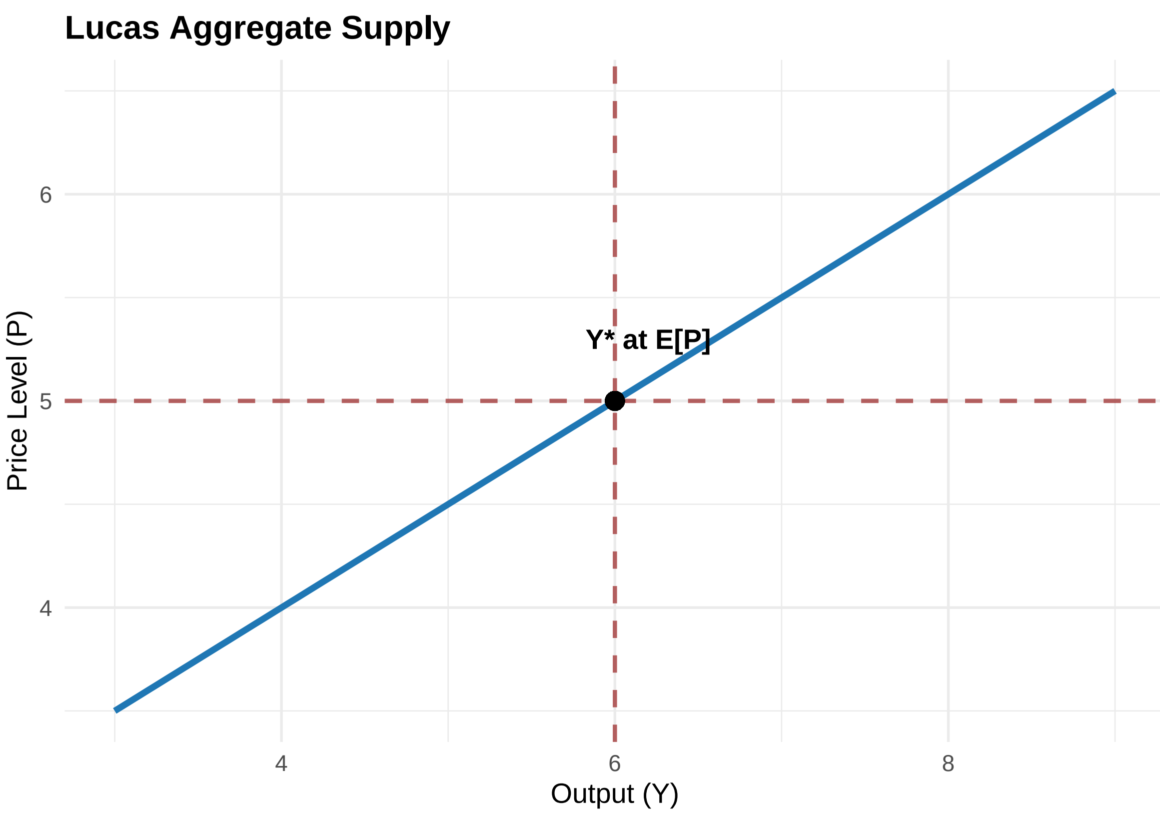

Figure 1: Lucas Aggregate Supply: $Y_t - Y^* = \kappa(P_t - E_{t-1}P_t)$

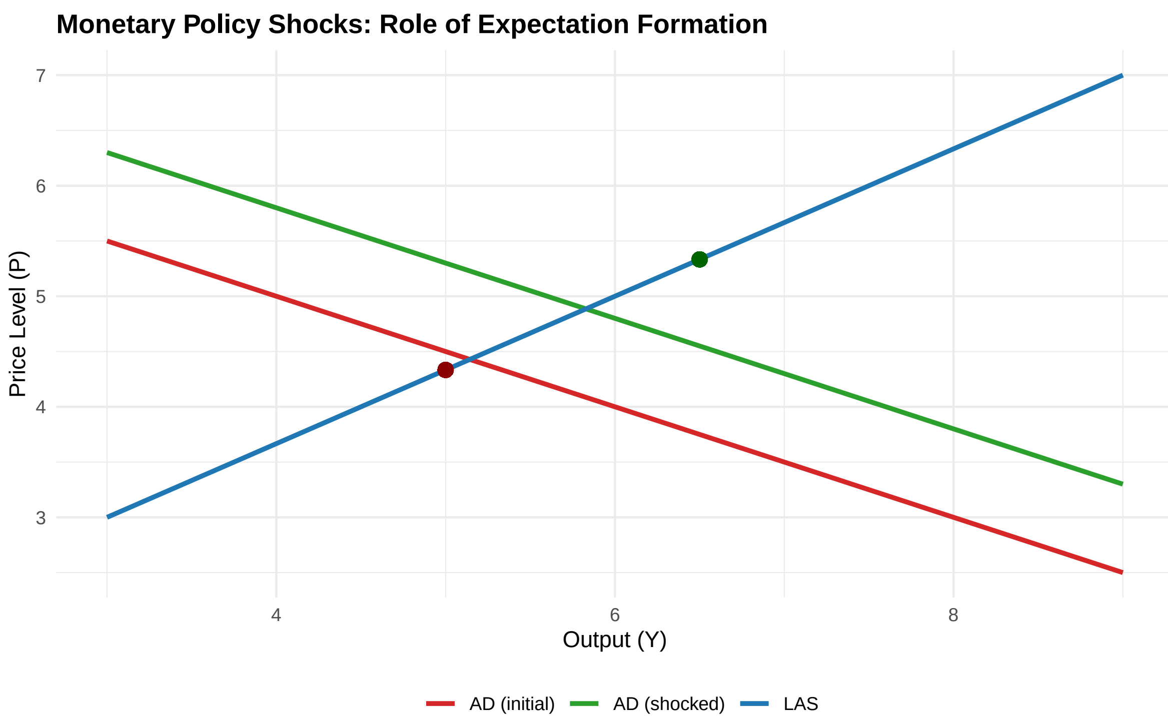

Figure 2: Policy predictability determines whether output effects are temporary.

Figure 3. Policy volatility reduces supply responsiveness via information effects. Greater policy volatility lowers the informational content of observed price movements. Agents cannot distinguish relative-price signals from aggregate noise, reducing supply response (lower κ) and steepening LAS.

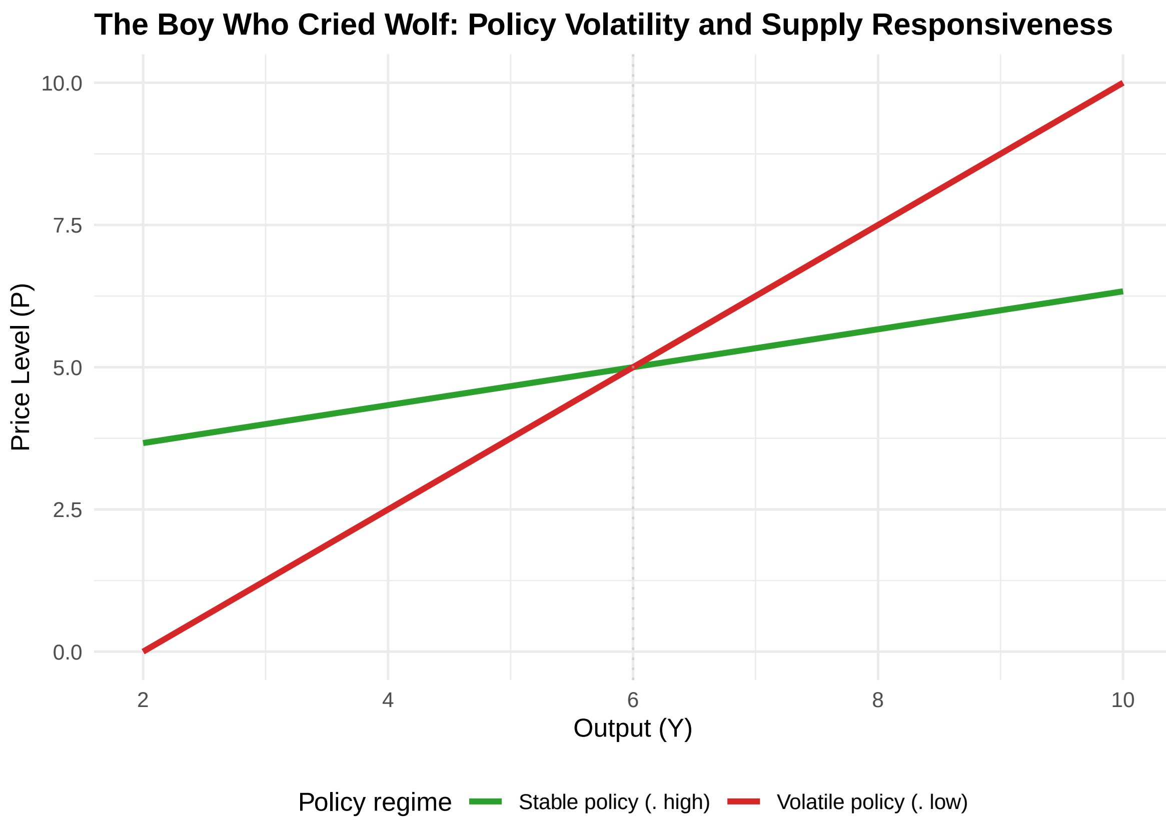

Figure 3: Higher policy volatility steepens Lucas supply through information effects.

7 Problem questions

1. In the LAS equation $Y_t - Y^* = \kappa(P_t - E_{t-1}P_t)$, interpret every term economically. Why does the model predict $E_{t-1}Y_t = Y^*$ under rational expectations?
2. Show that equilibrium output deviations are $Y_t - Y^* = (\kappa/(1+\kappa))(M_t - E_{t-1}M_t)$. What happens as $\kappa \to 0$? As $\kappa \to \infty$?
3. Explain why "random policy can move output" is not an argument for deliberate policy randomness. Explain using both (i) sign symmetry of shocks and (ii) the "boy who cried wolf" information mechanism.

4. Describe the asymmetric-information exception: what must be true about the central bank's information set relative to the private sector's for systematic stabilisation to operate under rational expectations?
5. Explain Fischer's wage-contract critique: why can anticipated monetary policy offset demand shocks when only part of the economy's wages are flexible?
6. Using high-frequency monetary policy shock data, explain how researchers operationalise "policy surprises." Why is this relevant to the New Classical policy-ineffectiveness result?

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Time-inconsistency literature: policy is strategic interaction with rational agents. Discretionary optimisation can backfire because promises today are not optimal tomorrow (after expectations set). Central to rules-versus-discretion analysis.

12 Modern evidence and real-world examples

12.1 Measuring policy surprises

In the New Classical model, the disturbance term is $M_t - E_{t-1}M_t$. Modern empirical work isolates *surprise components* of policy announcements using high-frequency financial market data.

A recent IMF working paper constructs a dataset of monetary policy shocks for 21 advanced economies and 8 emerging markets (2000–2022) using daily changes in interest rate swap rates around central bank announcements to identify unexpected shocks to the policy path. This maps directly onto the model's object: an unexpected policy shock is the empirical analogue of u_t in the money-growth process.

12.2 Credibility and anchored expectations

New Classical analysis makes credibility and expectations central. Modern central banks explicitly frame policy as expectations management.

The Federal Reserve explains that publishing its "Statement on Longer-Run Goals and Monetary Policy Strategy" helps the public understand the policy framework, which matters for transparency, accountability, and effectiveness.

The ECB's 2021 strategy statement emphasises a symmetric 2% medium-term inflation aim and explicitly notes that the effective lower bound can require "especially forceful or persistent" measures, potentially including a transitory period of inflation moderately above target.

IMF research provides evidence that better-anchored expectations are associated with less persistent inflation responses to shocks, reinforcing the New Classical emphasis on credibility and expectations formation.

12.3 Policy regime episodes

The Bank of Japan reference timeline documents major unconventional regimes from the late 1990s onward (ZIRP from 1999, QE from 2001, and later phases). It explicitly lists policy duration effect (forward guidance) as part of ZIRP—an expectations-management concept compatible with New Classical thinking.

Japan's 2024 framework change statement explains that QQE with yield curve control and negative rates were judged to have fulfilled their roles, with a shift back to guiding the short-term interest rate as the primary tool.

For the euro area, the ECB's June 2014 press release introducing a negative deposit facility rate provides a canonical policy regime expansion example: the ECB cut the deposit facility rate to -0.10% effective June 11, 2014.

During COVID-19, emergency actions show how central banks reacted to extraordinary shocks with large-scale measures that were partly about market functioning and partly about macro stabilisation.

13 Problem questions

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